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A star-shaped solid composite electrolyte containing multifunctional moieties with enhanced electrochemical properties for all solid-state lithium batteries

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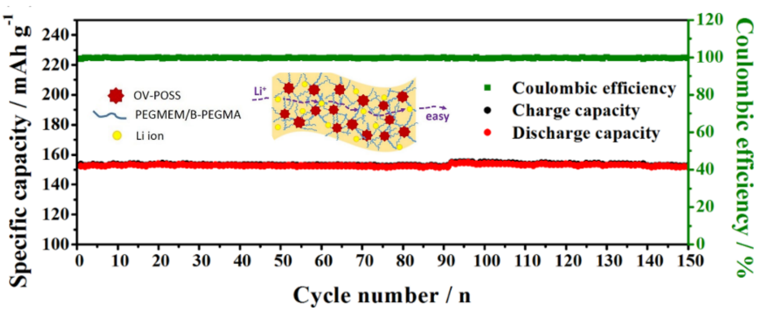
Abstract

Limited ionic conduction and poor solid/solid interfacial stability are crucial characteristics that impede the practical application of solid polymeric electrolytes. Herein, a star-shaped solid composite electrolyte (SCE) containing multifunctional components, including anion-trapping boron moieties (B-PEGMA), poly(ethylene glycol)methyl ether methacrylate (PEGMEM) and octavinyl octasilsesquioxane (OV-POSS) nanofiller, was developed via a simple free radical polymerization method. The unique star-shaped structure induced by OV-POSS is beneficial to increasing the movement of polymer chains and forming continuously interconnected ion-conducting channels and the boron moieties can promote lithium salt dissociation and increase the effective transmission of Li^+ in the electrolyte. This SCE exhibits an extremely high ionic conductivity of 3.44×10^{-4} S cm^{-1} and a high Li ion transference number of 0.58 at 25 ^{o}C, as well as excellent interfacial compatibility with the Li electrode leading to excellent rate performance and good cyclic stability in all-solid-state Li batteries.

Graphical abstract:



Keywords

All solid-state lithium batteries; Star-shaped polymer composite electrolyte; Poly (ethylene oxide); octavinyl octasilsesquioxane (OV-POSS); B-PEGMA.

1. Introduction

Solid polymer electrolytes (SPEs), with their light weight, easy film-forming ability and good viscoelasticity, show great potential for improving the energy density, operating temperature range, cycle life and structure flexibility of lithium ion batteries (LIBs) [1-4]. However, SPEs exhibit extremely low ionic conductivities at room temperature ($<10^{-6}$ S cm^{-1}), far lower than the conductivities of traditional liquid organic electrolytes [5, 6]. Furthermore, the interfaces between SPEs and the electrodes are non-fluidic solid/solid contacts and, as such, SPEs can not fully infiltrate the electrode materials like liquid electrolytes, resulting in degradation of cycling performance and successive capacity fading [7, 8]. Consequently, designing a solid electrolyte with high ionic conductivity and good electrode/electrolyte interface contact is a key challenge facing the development of high performance LIBs[9, 10].

Solid composite electrolytes (SCEs) with multicomponents and unique micro-nano structures for LIBs have attracted intensive attention recently [11, 12]. Grafting/blending nanofillers, such as ceramic particles [13-17], surface-modified nanoparticles [18-20] or carbonous nanomaterials [21, 22], has been considered as an efficient method to improve the electrochemical properties of SCEs. Among these SCEs, polyhedral oligomeric silsesquioxane (POSS), incorporated in the cage-like inorganic core formed by Si-O-Si bonds and organic functional groups surrounding the inorganic core, has been used as an effective nanofiller to improve ion conductivity and enhance mechanical strength due to its well-defined nanoscale organic/inorganic hybrid structure [11, 23-26]. Recently, our group demonstrated organic/inorganic nanocomposites comprised of OV-POSS and PEGMEM as SCEs for high performance LIBs [27]. Such SCEs with high ionic conductivity (1.13×10^{-4} S cm^{-1} at 25 ^{o}C) and improved Li transference number (0.35 at 25 ^{o}C) were found to be promising candidates for advanced electrolytes in high performance LIBs. It is suggested that appropriate compositional manipulation could be effective to facilitate higher ionic conductivities and, more importantly, a higher Li transference number that is another essential aspect to obtain good cycle performance for LIBs.

In search of SCEs with high ionic conductivity and high Li ion transference number, we have synthesized in this study a novel organic monomer of B-PEGMA containing cyclic borate ester groups and ether oxygen units. A star-shaped SCE of POSS-g-PEGMEM/B-PEGMA is then developed by grafting PEGMEM and B-PEGMA segments onto the OV-POSS moieties using a simple free radical polymerization method. It is well recognized that B atoms can coordinatively interact with anions of Li salt, since B is sp^2 hybridized and has an empty p-orbital [28, 29], and B-containing moieties can be used as an effective additive to promote the dissociation of lithium salts [30-32]. The star-shaped microstructure generated by

OV-POSS is favorable to enhance the movement of polymer chains that facilitate Li ion conduction and the cyclic borate ester groups in B-PEGMA segments may promote salt dissociation. A combination of these multifunctional components is an effective structural design to improve ionic conductivity and the Li ion transference number of the composite electrolyte simultaneously. Moreover, stable interfacial behavior and superior cycling and rate performance are observed for all solid-state LIBs assembled using this composite electrolyte.

2. Experimental section

2.1 Materials

Poly (ethylene glycol) dimethacrylate (PEGMA, M_n = 475 g mol^{-1}) and poly (ethylene glycol) methyl ether methacrylate (PEGMEM, M_n = 950 g mol^{-1}), obtained from Sigma-Aldrich, were used as received. Trimethyl borate (TMB), octavinyl octasilsesquioxane (OV-POSS, M_n = 633 g mol^{-1}), 2,2-azobisisobutyronitrile (AIBN), 2,5-dimethylhexane-2,5-diol and anhydrous acetonitrile were purchased from Aladdin and used as received. Lithium trifluoromethanesulfonate (LiTFSA) was purchased from Aladdin and dried at 120 °C for 10 h under vacuum conditions before use. Tetrahydrofuran (THF) and petroleum ether were purchased from Sinopharm Chemical Reagent Co., Ltd. and used as received.

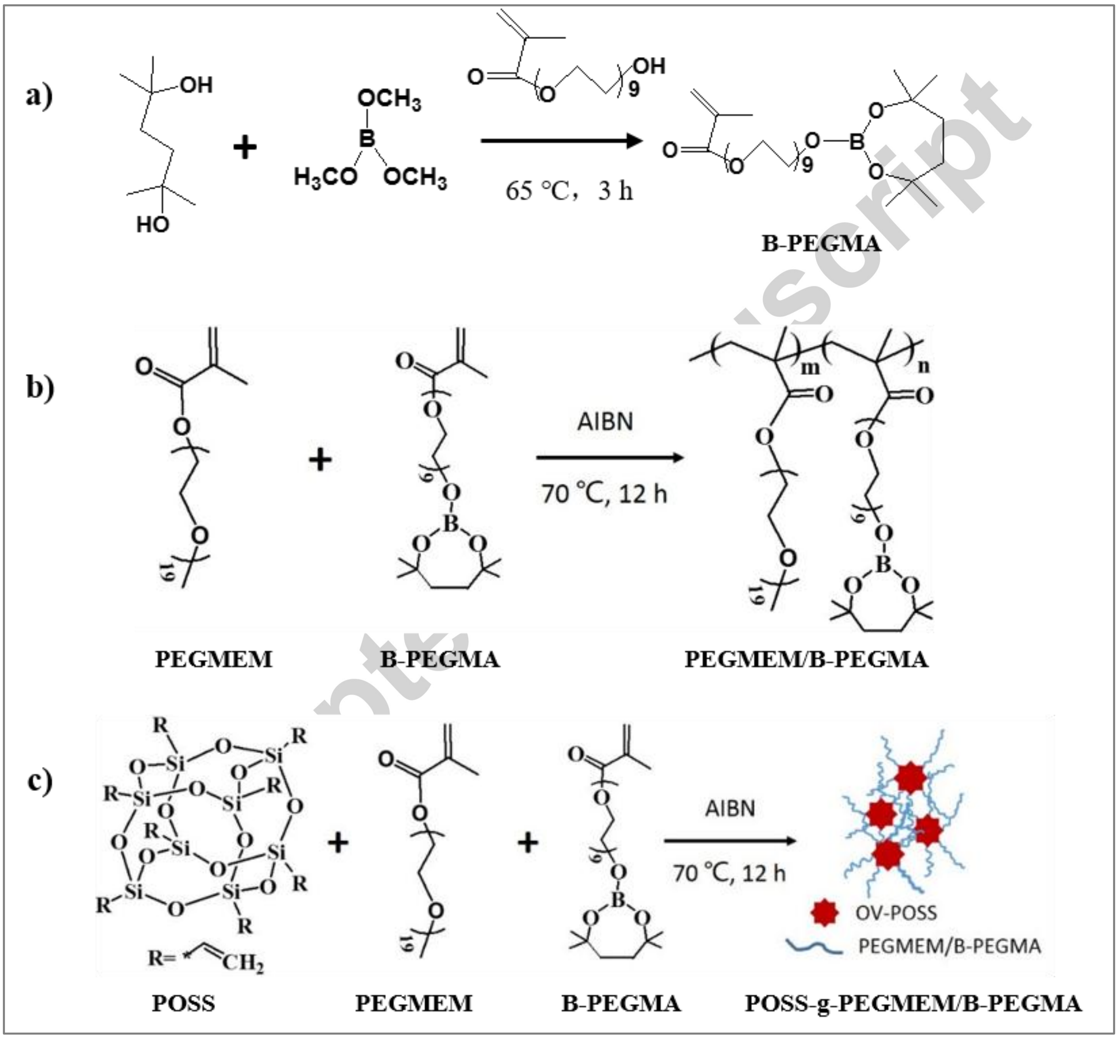
2.2 Synthesis of B-PEGMA monomer

The B-PEGMA monomer was synthesized as shown in **Scheme 1a**. 2,5-dimethylhexane-2,5-diol (1.0 g) and trimethyl borate (TMB) (0.9 mL), measured by a pipette, were dissolved in 50 mL anhydrous acetonitrile in a 100 mL three-neck flask. The solution was heated to 65 °C in oil bath and stirred for 1 h under an Ar atmosphere. Then PEGMA (2.4 g) was added to the three-neck flask and stirred for 3 h. After the reaction, the residual acetonitrile was removed by reduced pressure distillation at room temperature. The obtained product was dissolved in toluene and then the insoluble impurity was filtered out. The residual toluene was also removed by reduced pressure distillation at 65 °C. To obtain the B-PEGMA monomer, the product was dried under vacuum conditions at room temperature for 24 h.

2.3 Preparation of PEGMEM/B-PEGMA and POSS-g-PEGMEM/B-PEGMA SCEs

The PEGMEM/B-PEGMA and POSS-g-PEGMEM/B-PEGMA copolymers were synthesized via the free radical polymerization method, as shown in **Scheme 1b-1c**. Various stoichiometric mass ratios of B-PEGMA, PEGMEM and OV-POSS monomers were dissolved in acetonitrile in a 100 mL three-neck flask with AIBN as the initiator (0.5 wt% of total mass). The mixed solutions were heated to 70 °C for 12 h under an Ar atmosphere. The mixtures after the reaction were dissolved in THF and precipitated in petroleum ether three times to remove the residual monomer. The B-PEGMA/PEGMEM and POSS-g-PEGMEM/B-PEGMA copolymers were obtained after the products were dried at 60 °C under vacuum conditions for 12 h.

PEGMEM/B-PEGMA copolymer and LiTFSI, with the desired [Li]/[EO] molar ratio, were dissolved in THF and then dripped on a cellulose matrix, which was used to ensure dimensional stability and to improve the mechanical properties of the electrolyte [33]. The as-prepared composite electrolytes were dried in a vacuum oven at 70 °C for 12 h to remove the residual THF. POSS-g-PEGMEM/B-PEGMA copolymer and LiTFSA, with the desired [Li]/[EO] molar ratio, were dissolved in THF and then poured into a grooved Teflon mold. The mixture was put aside for 12 h at room temperature and then transferred to a vacuum oven at 100 °C for 12 h to remove the residual THF. The obtained SCEs were stored in a glove box for later application.



Scheme 1. a) Synthesis of B-PEGMA monomer； b) synthesis of the PEGMEM/B-PEGMA copolymer; c) synthesis of the POSS-g-PEGMEM/B-PEGMA copolymer.

2.4 Materials characterization and electrochemical measurements

^{1}H nuclear magnetic reso nance spectra (Avance III 400 MHz Digital NMR spectrometer) and Fourier transform infrared spectra (Nicolet 6700 FTIR spectrometer over the range of 4000-400 cm^{-1}) were used to characterize structure information for the B-PEGMA monomer and PEGMEM/B-PEGMA and POSS-g-PEGMEM/B-PEGMA copolymers. The thermal properties of the copolymers and composite electrolytes were evaluated by thermogravimetric analysis (TGA, SDTQ-600 TA), with a heating rate of 10 °C min^{-1} from 30 to 650 °C under a N_2 atmosphere, and differential scanning calorimetry (DSC, SDTQ-600 TA) with a heating rate of 10 °C min^{-1} from -100 to 80 °C under N_2 flow.

The ionic conductivity of the SCEs was measured with a PARSTAT 4000 system (Ametek Advanced Measurement Technology INC) from 25 to 100 °C over a frequency range of 0.1–100 kHz with a perturbation voltage of 10 mV. The samples were sandwiched between two stainless steel discs (d = 16 mm). The electrochemical stability was evaluated with a stainless steel/SCE/Li coin cell using linear sweep voltammetry (LSV) in a range from 2.5 to 7 V (versus Li/Li^+) at a scan rate of 10 mV s^{-1} at 25 °C. The direct current (DC) polarization/alternating current (AC) impedance method was employed to evaluate the lithium ion transference number (t^+) in a Li/SCE/Li symmetric coin cell. The Li/SCE/Li symmetric coin cell was also used to evaluate the interface stability between the Li electrode and the SCEs. To investigate the electrochemical performance of the SCEs in LIBs, $LiFePO_4$ (LFP)/Li coin cells containing a Li anode, a $LiFePO_4$ cathode and the SCEs were assembled in an Ar filled glove box. The mixture of 80 wt% $LiFePO_4$, 10 wt% carbon black and 10 wt% PVDF dispersed in N-methyl-2pyrrolidone (NMP) was coated on an aluminum foil, and then dried under vacuum at 110 ^oC for 12 h to remove the residual NMP. The diameter of Li metal electrode, $LiFePO_4$ cathode and electrolyte membrane in the coin cell is 16, 12 and 18 mm, respectively. The charge/discharge tests were performed in a battery test system (LANHE CT2001A, Wuhan LAND electronics Co., PR China) between 2.5 and 3.7 V at different C-rates and temperatures.

3. Results and discussion

3.1 Structural characterization

Structural information for B-PEGMA, PEGMEM/B-PEGMA and POSS-g-PEGMEM/B-PEGMA was determined by ^1H NMR analysis (**Fig. 1**) and FTIR spectroscopy (**Fig. 2**). As shown in **Fig. 1a**, the resonance peaks at 1.89 ppm (b), 5.52 ppm (h) and 6.08 ppm (g) are assigned to the protons from the PEGMA backbone. The protons of the CH_2–CH_2–O units from the PEGMA segments are observed at 3.52-4.24 ppm (c) [23]. The resonance peaks at 1.16 ppm (d) and 1.51 ppm (e) are assigned to the protons of alkyl and ethylene groups of the boronic ester, indicating that the B moiety was successfully grafted onto the PEGMA segments [34]. The peaks at 2.32 ppm (a) and 0.89 ppm (b) in **Fig. 1b** are attributed to the protons of the PEGMEM backbone and the resonance peak at 3.39 ppm (f) is assigned to the protons of terminal CH_3 from the PEGMEM segments. The multiple peaks observed at 5.52–6.08 ppm (g and h) in **Fig. 1c** are the unreacted vinyl protons of POSS, verifying that eight vinyls of POSS are not fully involved in the reaction [35].

The FTIR spectra for B-PEGMA, PEGMEM/B-PEGMA(5/5) and POSS-g-PEGMEM/B-PEGMA(5% POSS) are presented in **Fig. 1d**. For B-PEGMA monomer, the vibrational peak at 668 cm^{-1} is assigned to the B-O bonds, indicating that the B moiety has been successfully grafted onto the PEGMA units [34]. The strong peak at 1728 cm^{-1} is attributed to C=O bonds and the absorption peak at 1627 cm^{-1} can be ascribed to the C=C bonds. After free-radical polymerization, the peak intensities of the C=C bonds show only slight change since eight vinyls of OV-POSS do not fully participate in the reaction. The strong peak at 1106 cm^{-1} is attributed to C-O-C bonds, which coincides with the absorption peak of Si-O-Si bonds in OV-POSS macromonomer. And thus the peak intensities of POSS-g-PEGMEM/B-PEGMA at 1106 cm^{-1} increase a lot.

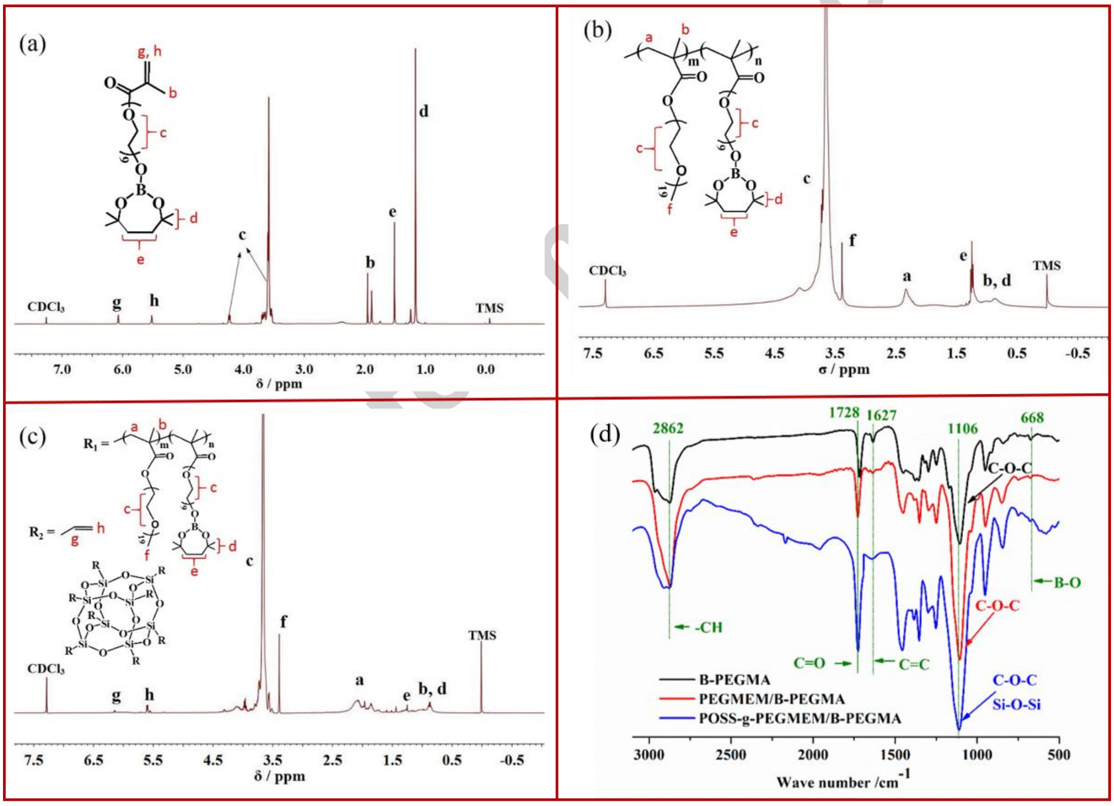


Fig. 1 ^{1}H NMR spectra (a-c) and FTIR spectra (d) for B-PEGMA (a), PEGMEM/B-PEGMA(5/5) (b) and POSS-g-PEGMEM/B-PEGMA(5 wt% POSS) (c).

3.2 Ionic conductivity

Fig. 2a shows the temperature-dependent ionic conductivity of PEGMEM/B-PEGMA (5:5) SCE with various LiTFSA concentrations. The conductivity first increases with the addition of LiTFSA, up to a maximum value corresponding to a [Li]/[EO] molar ratio of 1/20, and then decreases with a further increase in LiTFSA concentration. This phenomenon, which was commonly observed in previous studies [27, 33, 36], involves a trade-off between the mobility of the EO chains and the number of charge carriers. The optimum [Li]/[EO] molar ratio corresponding to the highest ionic conductivity is 1/20 in this research work. **Fig. 2b** shows the temperature-dependent ionic conductivity of PEGMEM/B-PEGMA SCEs with various B-PEGMA contents. Ionic conductivity increases as temperature rises due to the reduced crystallinity of the EO units and the enhanced chain mobility of the polymer matrix. B-PEGMA content also has a slight effect on the conductivity of the SCEs. As shown in **Fig. 2c**, PEGMEM/B-PEGMA SCE exhibits its highest conductivity of 5.37×10^{-5} S cm^{-1} at 25 ^{o}C at a B-PEGMA content of 50 wt%. However, the strong aggregation tendency of excess B-PEGMA segments could increase the orderliness of EO segments since B-PEGMA segments also contain EO groups, resulting in the decrease of the free mobility of EO segments and accompanying reduction of the ionic conductivity. To further improve conductivity of the electrolyte, PEGMEM, B-PEGMA and OV-POSS monomers with eight functional corner groups were selected for the synthesis of star-shaped copolymers, where the PEGMEM/B-PEGMA weight ratio is 5:5. **Fig. 2d** shows the temperature-dependent ionic conductivity of star-shaped POSS-g-PEGMEM/B-PEGMA SCEs with various POSS contents. When the OV-POSS content is low, it is apparent that the introduction of the OV-POSS macromonomer significantly improves the conductivity by disrupting the order of EO segments and increasing the free volume for the motion of lithium ion. Simultaneously, additional segmental motion of EO chains grafted on OV-POSS particles enable low-energy paths along the nanoparticles/polymer matrix interface to transport lithium ions, as illustrated in **Fig. 3b**. The ionic conductivity reaches a highest value of 3.44×10^{-4}, 1.09×10^{-3} and 1.61×10^{-3} S cm^{-1} at 25, 60 and 80 ^{o}C respectively, when the POSS content is 5 wt%. These values are much higher than those for PEGMEM/B-PEGMA SCEs and other PEO-based SCEs [27, 33, 37-41], as tabulated in **Table S1**. However, with the further increase of OV-POSS, the ionic conductivity decreases since the bulky OV-POSS groups cannot provide channels for lithium ion transport and the excess OV-POSS nanoparticles will reduce the free volume of the polymers and reduce chain mobility [34].

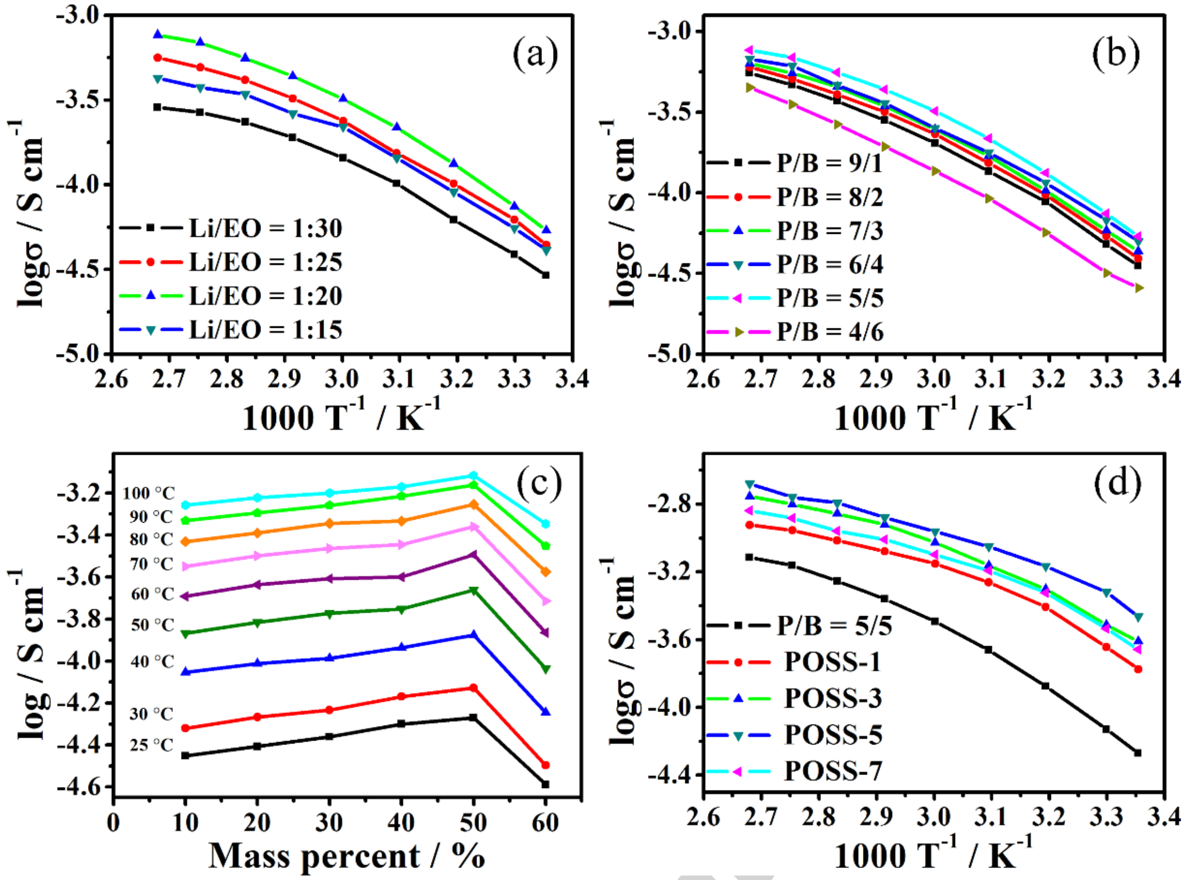


Fig. 2 (a) Temperature-dependent ionic conductivity for PEGMEM/B-PEGMA (5:5) SCEs with various LiTFSA concentrations; (b) temperature-dependent ionic conductivity of PEGMEM/B-PEGMA SCEs with different B-PEGMA contents; (c) ionic conductivity of PEGMEM/B-PEGMA SCEs with various weight fractions of B-PEGMA at different temperature; (d) temperature-dependent ionic conductivity of POSS-g-PEGMEM/B-PEGMA star-shaped SCEs.

The improvement in ionic conductivity for star-shaped POSS-g-PEGMEM/B-PEGMA SCEs can be ascribed to the reduced crystallinity of the EO groups. DSC analysis of PEGMEM/B-PEGMA and POSS-g-PEGMEM/B-PEGMA SCEs was carried out, as shown in **Fig. 3a**. Thermal parameters, such as glass transition temperature (T_g), melting temperature (T_m) and melting enthalpy (ΔH_m), are summarized in **Table 1**. In contrast to the PEGMEM/B-PEGMA copolymer, the star-shaped POSS-g-PEGMEM/B-PEGMA copolymer has lower T_g, T_m and ΔH_m values, suggesting that the addition of POSS increase the amorphous fraction of the copolymer, which is beneficial for ion transport. Besides, our previous study has showed that the glass transition temperature of EO groups in star-shaped copolymers is lower than that of the corresponding linear copolymers when they have the same POSS content [27], demonstrating that the unique star-shaped structure also facilitates the reduction of the crystallinity of EO segments. Simultaneously, the steric hindrance effect among POSS groups offers additional free volume for the motion of EO chains and the star-shaped arms of EO segments on the surface of the POSS particles provide low energy, fast and continuously interconnected ion-conducting channels, resulting in higher ionic transport efficiency, as schematically illustrated in **Fig. 3b**.

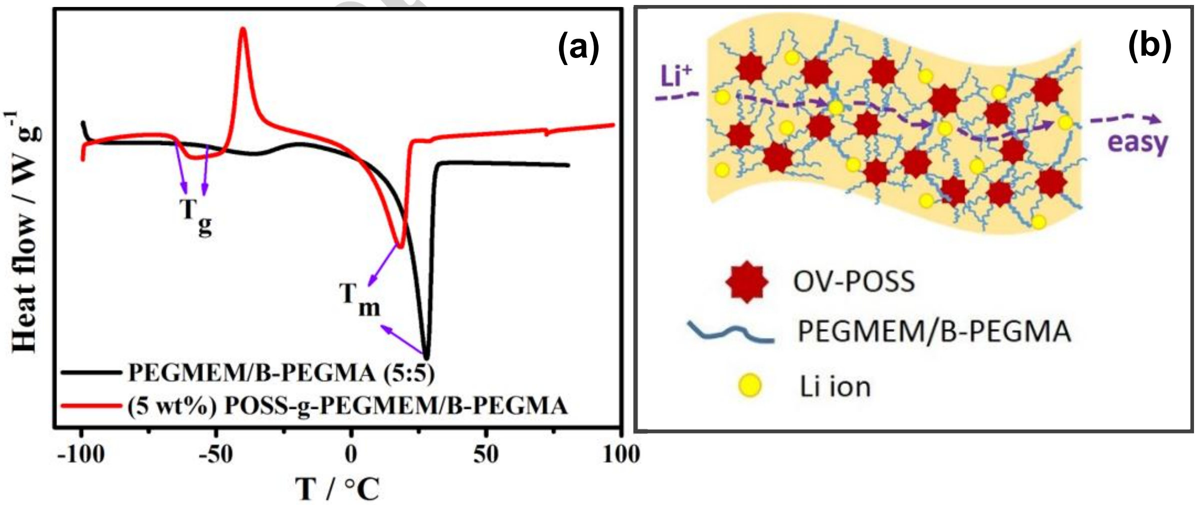


Fig. 3 (a) DSC analysis of PEGMEM/B-PEGMA (5:5) and (5 wt%) POSS-g-PEGMEM/B-PEGMA copolymer; (b) illustration of Li^+ transport mechanism

in POSS-g-PEGMEM/B-PEGMA star-shaped SCE.

Table 1 Thermal data for PEGMEM/B-PEGMA (5:5) and (5 wt%) POSS-g-PEGMEM/B-PEGMA polymer matrix

Sample	T_g (^{o}C)	T_m (^{o}C)	ΔH_m (J g^{-1})
PEGMEM/B-PEGMA (5:5)	-50.29	27.78	61.25
(5 wt%) POSS-g-PEGMEM/B-PEGMA	-62.42	17.86	43.60

3.3 Thermal and electrochemical stabilities

Thermal and electrochemical stabilities are two crucial assessment parameters closely related to battery safety performance. **Fig. 4a** is the thermogravimetric curve for POSS-g-PEGMEM/B-PEGMA SCE; there was no obvious weight loss below 190 ^{o}C. The weight loss between 190 and 385 ^{o}C can be assigned to the thermal decomposition of the polymer matrix. The second stage of weight loss begins at 385 °C and can be attributed to the decomposition of LiTFSA and Si-O-Si segments, implying that the POSS-g-PEGMEM/B-PEGMA SCE has good thermal stability below 190 ^{o}C. The inset in **Fig. 4a** is a photograph of POSS-g-PEGMEM/B-PEGMA SCE, which is free standing and displays good dimensional stability. The as-prepared SCE exhibits smooth surface and the thickness of the membrane is around 220 μm (**Fig. S1).** The electrochemical stability was measured using cyclic voltammetry scan at 60 ^{o}C (**Fig. 4b**). The negative scan was swept from 2.5 to -0.5 V and then back to 2.5 V and the positive scan was from 2.5 to 7 V at a sweep rate of 5 mV s^{-1}. The inset is the enlarger part in the potential range of 5 to 6.5 V. Lithium plating peak appears at -0.381V and the corresponding stripping peak occurs at 0.138 V. Besides, broad reversible peaks shown as red arrows in **Fig. 4b** are visible, which could be ascribed to the reaction of the TFSI$^-$ anions and impurities [33, 42]. The intersection of the tangent line before and after the voltage change, 5.84 V, represents the significant decomposition voltage of the electrolyte, indicating that these SCEs exhibit outstanding electrochemical stability and have a great promise for application in high voltage LIBs.

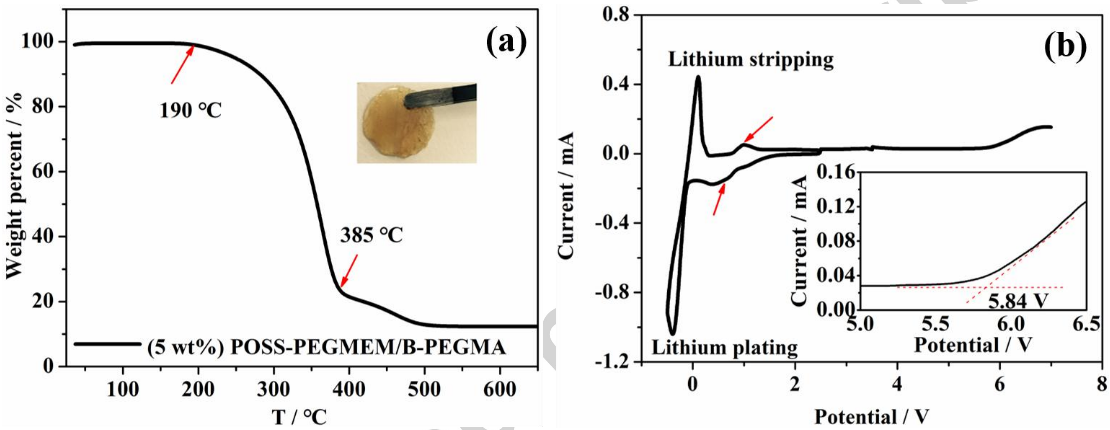


Fig. 4 TG curve (a) and cyclic voltammograms (b) for POSS-g-PEGMEM/B-PEGMA SCE. The inset in (a) is a photograph of POSS-g-PEGMEM/B-PEGMA SCE and the inset in (b) is the enlarger version in the potential range of 5 to 6.5 V.

3.4 Li$^+$ transference number

The Li$^+$ transference number (t^+) reflects the proportion of Li$^+$ in all charge carriers and is an important parameter for characterizing the contribution of Li$^+$ to ion conduction [43]. t^+ was calculated using the following equation (**Eq.1**)

$$t^+ = \frac{I_S(V - I_0 R_0)}{I_0(V - I_S R_S)} \quad (1)$$

Fig. 5a shows the DC polarization curve for star-shaped

POSS-g-PEGMEM/B-PEGMA SCE with initial and steady-state currents of I_0 = 26.8 μA and I_s = 16.49 μA, respectively. **Fig. 5b** is the AC impedance spectra before and after polarization and the inset shows the equivalent circuit used to fit R_0 and R_s. R_0 (57.14 Ω) and R_s (58.17 Ω) are the initial and steady-state interfacial resistances between the electrolyte and the Li electrode. The value of t^+ for the star-shaped POSS-g-PEGMEM/B-PEGMA SCE is 0.58, which is slightly higher than that of PEGMEM/B-PEGMA (5:5) electrolyte of 0.56 **(Fig.S2)**, demonstrating a small increase of Li^+ transference number by the addition of POSS moieties. However, the value of t^+ for POSS-g-PEGMEM/B-PEGMA SCE is much higher than that of POSS-g-PEGMEM SCE reported in our previous work [27] and other PEO-based SCEs [27, 38, 39, 41], as shown in **Table S1**. The results indicate that the borate groups in the B-PEGMA segments have a substantial effect on increasing the effective transport of Li^+ in the electrolyte, since the B atoms interact in coordination with anions of the Li salt and promote the dissociation of Li salt. A high Li^+ transference number is beneficial in reducing concentration polarization and enhancing the charge/discharge capacity of LIBs [44].

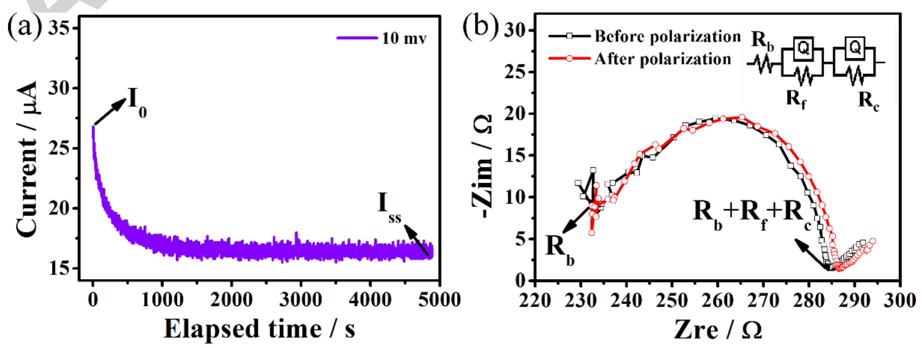


Fig. 5 Chronoamperometry profile (a) and AC impedance spectra before and after

polarization （b）for a symmetric Li/Li coin-type cell with

POSS-g-PEGMEM/B-PEGMA SCE at 25 ^{o}C.

3.5 Interfacial stability of SCE/Li interface and battery performance

Polarization testing for a Li/Li symmetric battery was carried out under constant current densities of 0.05 and 0.1 mA cm^{-2} to investigate the interfacial stability between POSS-g-PEGMEM/B-PEGMA SCE and the Li electrode. The differences in voltage caused by constant charge/discharge current density are closely related to the interfacial resistance [45, 46]. As shown in **Fig. 6a**, POSS-g-PEGMEM/B-PEGMA SCE exhibits polarization voltages of ~0.27 V at 0.05 mA cm^{-2} and ~0.48 V at 0.1 mA cm^{-2}, which are significantly lower than those of POSS-g-PEGMEM SCE (0.30 ~ 0.54 V at 0.05 mA cm^{-2} and 0.78 ~ 0.96 V at 0.1 mA cm^{-2} in **Fig. S3**). Moreover, polarization curves of POSS-g-PEGMEM/B-PEGMA SCE are regular while the polarization curve of POSS-g-PEGMEM SCE exhibits obvious random fluctuation, indicating that the introduction of B-PEGMA moieties help to promote uniform Li deposition/striping and improve the interfacial stability of composite electrolyte/Li interface, and showing potential for application in high performance LIBs.

LFP/Li coin cells using POSS-g-PEGMEM/B-PEGMA as the electrolyte were assembled to investigate the battery performance of the SCE. **Fig. 6b** presents the discharge curves for LFP/Li battery at 60 ^{o}C under various C rates with a charge current density of 0.1 C. Discharge capacities of more than 150 mAh g^{-1} were achieved at 0.1, 0.2, 0.5 and 1 C. When the discharge current density was increased to 2 and 4 C, the capacities were 140.1 and 105 mAh g^{-1}, respectively, and flat potential plateaus were still be observed at 2 C. The superior rate performance is closely related to the high ionic conductivity and Li ion transference number and especially the stable interfacial behavior between electrolyte and electrode materials.

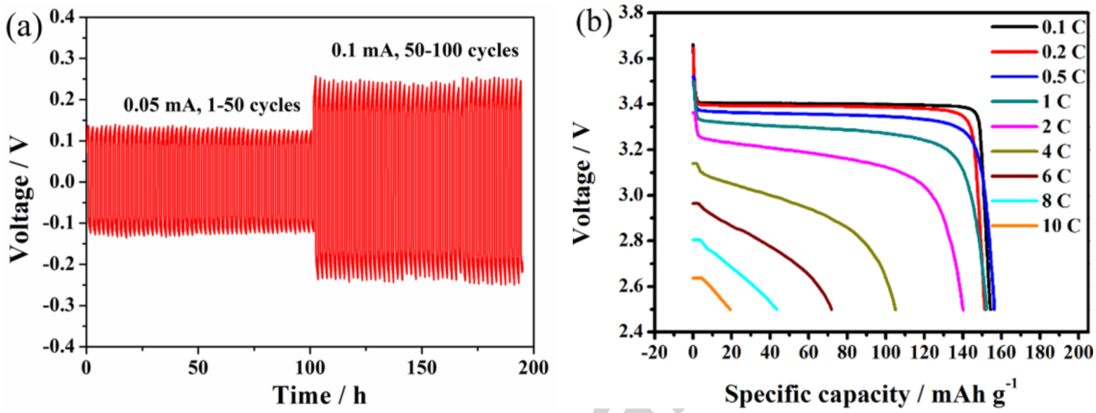


Fig. 6 (a) Polarization test for Li/Li symmetric battery using POSS-g-PEGMEM/B-PEGMA SCE at 25 °C; (b) discharge curves for LFP/Li battery using POSS-g-PEGMEM/B-PEGMA SCE at 60 °C under various C-rates with a charge current density of 0.1 C.

Long-term cycling stability of LFP/Li coin cells using POSS-g-PEGMEM/B-PEGMA as the electrolyte was also studied at a current density of 0.5 C at different temperatures, as shown in **Fig. 7**. The cell delivers an initial discharge capacity of 131.1 mAh g^{-1} and retains 120.8 mAh g^{-1} after 100 cycles at 25 °C. However, fluctuations in discharge capacity and coulombic efficiency are noticeable during the cycle process, indicative of a relatively poor Li ion extraction/insertion process at low temperature. When the temperature was increased to 60 °C, the cell exhibits an initial discharge capacity of 152.3 mAh g^{-1} and nearly no capacity fading after 150 cycles, demonstrating that a highly reversible Li ion extraction/insertion process is obtained at 60 °C. Note that the capacity fluctuation during cycle 92 is caused by an abrupt power failure of the testing system. Such cells deliver higher initial discharge capacities at 25 and 60 °C than those assembled with other PEO-based SCEs [27, 38, 41, 47]，as shown in **Table S1**. Besides, AC impedance test for LFP/Li coin cells after different cycles at 25 and 60 ^{o}C was carried out to evaluate the interfacial properties during cycling process, as shown in **Fig. S4**. It is apparent that the interfacial resistance at 25 ^{o}C is three times of that at 60 ^{o}C, demonstrating improved interfacial stability of electrolyte/electrode interfaces at high temperature. The improved cyclability can be attributed to the high ionic conductivity and Li^+ transference number and especially the good interfacial stability between the electrolyte and electrode materials at high temperature.

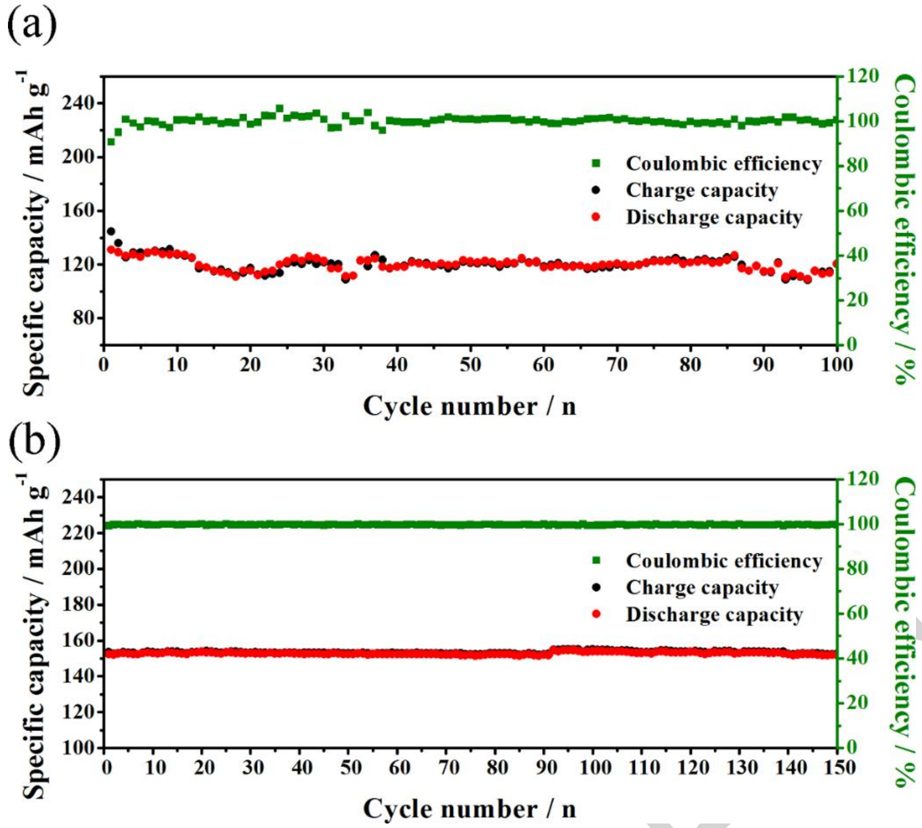


Fig. 7 Cycling performance of LFP/Li batteries assembled using POSS-g-PEGMEM/B-PEGMA SCE with a current density of 0.5 C at 25 °C (a) and 60 °C (b).

Conclusion

A star-shaped POSS-g-PEGMEM/B-PEGMA SCE with multifunctional components was developed through grafting PEGMEM and B-PEGMA segments onto OV-POSS groups using a simple free radical polymerization method. A high ionic conductivity of 3.44×10^{-4} S cm^{-1} and a high Li ion transference number of 0.58 at 25 °C are obtained, which can be ascribed to the star-shaped microstructure and the coordinated interaction between borate atoms and anions of the Li salt. The star-shaped POSS-g-PEGMEM/B-PEGMA SCE exhibits good interface stability with the Li anode material. LFP/Li coin cells using POSS-g-PEGMEM/B-PEGMA as the electrolyte demonstrate superior rate performance and long-term cycling stability at both room and elevated temperature.

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Supporting Information

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Highlights

- POSS nanoparticles offer interconnected ion-conducting channels.
- B atoms promote salt dissociation and increase the effective transmission of Li^+.
- The electrolyte exhibits a high ionic conductivity of 3.44×10^{-4} S cm^{-1} at 25 ^oC.
- The electrolyte exhibits a high lithium ion transference number of 0.58 at 25 ^oC.
- Excellent rate and cycle performance is obtained in solid-state Li-ion batteries.